

SUYU CHENG

(+1) 416-732-4796 | suyu0229@gmail.com | Toronto, ON, Canada
linkedin.com/in/suyu-cheng | github.com/Suyu0114

Data Engineering · Analytics Engineering · Business Intelligence

PROFESSIONAL SUMMARY

Software engineer with 6+ years building data-intensive systems for a multinational manufacturer — ERP/PLM data integration, SQL-heavy backends (MSSQL stored procedures), and analytics platforms used daily by executives, planners, and finance — as part of a 2–3 person internal development team. Recently completed postgraduate IT studies in Toronto (Honours) covering data warehousing, big data, and machine learning, and built three end-to-end Python + PostgreSQL analytics platforms with automated ETL, statistical modeling, and deployed web frontends. Authorized to work in Canada.

TECHNICAL SKILLS

- **Data Engineering:** SQL (MSSQL — 6 yrs production; PostgreSQL; Oracle PL/SQL), Python (pandas, statsmodels), ETL / data integration, data modeling, data warehousing, stored procedures, GitHub Actions scheduling
- **Business Intelligence:** Power BI, Tableau, Excel, dashboard design, KPI tracking
- **Programming:** C#, .NET Core 6, Vue.js 3
- **Tools & Platforms:** Git, Linux (Ubuntu), Jira, Trello, Agile / Scrum
- **Languages:** English (Professional), Mandarin (Native), Taiwanese (Native)

PROFESSIONAL EXPERIENCE

Seasonic Electronics — Taipei, Taiwan

Sep 2020 – Jun 2024

Senior Software Engineer

MRP Platform

- Designed and built a cross-system MRP data platform on **MSSQL / .NET Core 6**, integrating **ERP** (inventory, POs, purchase requests) and **PLM** (BOM, material master) into a unified data model that resolved material-code mismatches across **~1,000 raw material SKUs**, replacing manual Excel-based workflows used by production planning teams at HQ and factory sites of a power supply manufacturer.
- Implemented **BOM hierarchy traversal (C# recursion)** and **position-aware alternate-material substitution (MSSQL stored procedures)** to compute multi-level material shortages across one-stage (region-specific packaging) and two-stage (core product) BOMs on demand; **reduced material requirements planning cycle from 3 days to under 10 minutes per run** and codified previously tacit factory-floor substitution rules into structured data.

Fixed Asset Management System

- Built a centralized fixed-asset system (**MSSQL / .NET Core 6**) managing **~2,000–3,000 HQ assets** with a **lifecycle-state schema**, **ERP integration**, **QR-code lookup**, and auto-sync from the e-form approval workflow; **cut annual inventory data preparation from 5 workdays to under 0.5 day**.

E-signature Workflow System

- Architected a configurable **e-approval workflow engine (MSSQL / .NET Core 6)** serving **~150 users across Taiwan, Europe, the US, and China**: modeled the company org hierarchy (departments, reporting lines, deputies, approval levels) as relational data, with **stored procedures dynamically resolving multi-stage approval chains** per form type, applicant department, and **amount-based escalation rules**.
- Digitized **10+ paper approval processes** across finance, procurement, general affairs, HR, and engineering change management — supporting configurable sign-off quorums (all / any / designated approvers), ad-hoc approver insertion, per-stage email notifications, and **auto-sync of approved results into downstream systems** (e.g., the fixed-asset platform); enabled executives to approve remotely from mobile, eliminating multi-day waits on traveling CEO/Chairman sign-offs and replacing email-based proxy signing for overseas managers.

Digital Technical Documentation Platform

- Led a **three-person team** to build a technical-documentation platform for the Document Control Center — personally owning **schema design and stored procedures** — with **document-level access permissions** and **event-driven update notifications**, replacing Excel-and-paper document control.

ACTi Corporation — Taipei, Taiwan

Sep 2018 – Aug 2020

Sales Performance Forecasting / Dashboards / Commission

- Built a **win-rate-weighted sales forecasting system** (.NET Framework / MSSQL / jQuery / Vue.js 2) querying **live ERP and CRM databases**, with interactive drill-down from team-level rollups to individual project composition and linked CRM records; used directly by the **CEO for monthly, quarterly, and annual sales performance reviews**.
- Re-engineered a legacy, unmaintained **sales-commission engine** into maintainable **MSSQL stored procedures** handling multi-dimensional payout rules (**product model, country, order size, customer type**), and built detail views so sales staff could verify payout composition directly — a finance- and CEO-critical monthly process.
- Developed **company performance trend dashboards (Highcharts)** unifying CRM pipeline forecasts, projects, ERP orders, and shipments **across two database schemas** for integrated management analysis.

PROJECTS

World Cup 2026 Prediction & Analytics Platform — Python, PostgreSQL (Supabase), Next.js / TypeScript

github.com/Suyu0114/wcup2026-analytics | wcup2026-analytics.vercel.app/en

- End-to-end forecasting platform: idempotent, fail-loud ETL (fixtures, Elo ratings, betting odds with change-detection snapshots) → Dixon–Coles Poisson engine fitted on historical internationals behind a validation log-loss gate → Monte Carlo tournament simulation (group stage + knockout bracket) → bilingual web frontend presenting model-vs-market comparisons; recompute automated via GitHub Actions.

Toronto Blue Jays Analytics Site (live) — Python ETL, Supabase Postgres, Next.js, D3.js

bluejaysfanweb.vercel.app/en

- Daily scheduled ETL (GitHub Actions cron) ingesting MLB Stats API and Statcast into Postgres with idempotent backfills, powering a deployed bilingual site with spray charts, pitch heatmaps, WAR component breakdowns, and rolling-form analytics.

MLB Player-Performance Panel Study — Python, statsmodels / linearmodels

- Pre-registered hypothesis study across 1,181 players × 10 seasons on an 18-table PostgreSQL warehouse; OLS and PanelOLS (entity/time fixed effects) with BH-FDR multiple-comparison correction; honestly reported null result.

EDUCATION

Humber Polytechnic — Toronto, Canada

Sep 2024 – Jun 2026

Postgraduate Degree, Information Technology Solutions

Honours (3 of 4 terms). Relevant coursework: Big Data 1 (96) & 2 (90), Machine Learning (96), Deep Learning, Business Intelligence, Fundamentals of Data Analytics, Oracle Data Warehouse Fundamentals (91) & Implementation, Oracle DBA 1, Oracle Database Programming (PL/SQL), Intro to Database & SQL (90)

National Formosa University — Yunlin, Taiwan

Sep 2011 – Jun 2015

Bachelor of Business Administration

ADDITIONAL INFORMATION

Chairperson, Employee Welfare Committee, Seasonic Electronics

Sep 2022 – Jun 2024

Coordinated employee activities, vendor partnerships, and internal engagement initiatives — demonstrating leadership and cross-functional coordination.

Representative Powerlifting Athlete, Yilan County Citizens Sports Games

2022

Squat 205 kg / Bench Press 120 kg / Deadlift 230 kg.